DWG. No. 310SD03-743

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GENERATOR PROTECTION/BREAKER LOGIC DIAGRAM (STEAM TURBINE GENERATOR)

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REFERENCE DRAWING

1. ONE LINE DIAGRAM GENERATOR MAIN CIRCUIT
   HTC DWG. NO.: 310SD03-741
2. TURBINE GENERATOR INTERLOCK BLOCK DIAGRAM
   HTC DWG. NO.: 310SC90-774
3. ATS UNIT SEQUENCE CONTROL PROGRAM
   HTC DWG. NO.: 310SC90-775

NOTE:
SYNCHRONIZING OPERATION LOGIC IS WRITTEN BY CURRENT HTC UNDERSTANDING BASED ON MOM OF 17-19 JULY 2013 XSM-CO-PRJ-058 REV.0

三菱日立パワーシステムズ株式会社
MITSUBISHI HITACHI POWER SYSTEMS,LTD.
2015-06-11

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PROJECT TITLE / LOCATION

SMC DAVAO POWER PLANT PROJECT
MALITA, PROVINCE OF DAVAO DEL SUR, PHILIPPINES

Drawn : Y.Kawase	Name/Date	/2013-05-31
Checked : S.Mizukami	Name/Date	/2013-05-31
Approved : N.Miyamoto	Name/Date	/2013-05-31

Title,supplementary title

GENERATOR PROTECTION / BREAKER LOGIC DIAGRAM (STEAM TURBINE GENERATOR)

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

OWNER / ADDRESS

San Miguel Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550, METRO MANILA, PHILIPPINES

Responsible dep.
Hitachi Electrical Engineering Section / Electrical Department

Scale - : -

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CAD SOFTWARE	Microsoft Visio2010	SIZE DESIGNATION	A3

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SYMBOL LEGEND

SYMBOL	EXPLANATION
(N) COS	CHANGE OVER SWITCH (N) : LOCATION
(N) PB	PUSH BUTTON (N) : LOCATION
(A) (N) / (A) (N)	DETECTION OR EXPLANATION OF SIGNAL (A) : NAME OF EQUIPMENT (N) : LOCATION
(N) CS	CONTROL SWITCH (N) : LOCATION
A	"AND" GATE
OR	"OR" GATE
	"NOT" GATE
DI t	TIMER (TDPU : TIME DELAY PICK-UP) t : SETTING TIME
DO t	TIMER (TDDO : TIME DELAY DROPOUT) t : SETTING TIME
N	DEVICE (RELAY) N: DEVICE NO.
OR	CONTACT, DISCONNECTING SWITCH (DS) or EARTHING SWITCH (ES)
	CONTACTOR
	FUSE
	VOLTAGE TRANSFORMER (VT)
OR	TRANSFORMER
	CIRCUIT BREAKER (CB)
	SYNCHROSCOPE
V	VOLTMETER
F	FREQUENCY METER
	TERMINAL POINT
GEN	GENERATOR
R	RESISTOR
	LAMP INDICATION (R:RED, G:GREEN, W:WHITE, A:AMBER)

SYMBOL	EXPLANATION
C A OR A B B C	PHASE VECTOR
	GROUNDING TRANSFORMER
ANN	OUTPUT SIGNAL TO ALARM
SOE	OUTPUT SIGNAL TO SEQUENCE OF EVENT RECORDER (OTHERS)
FTRP	OUTPUT SIGNAL TO FULL TIME RECORDER PANEL (OTHERS)
A N	LOGIC CONTINUATION (DESTINATION) A : CONNECTION POINT N : SHEET No. SH-TA- : SHEET OF "STEAM TURBINE GENERATOR AUXILIARY INTERLOCK DIAGRAM" SH-G- : SHEET OF THIS DRAWING SH-* : SAME SHEET
A N	LOGIC CONTINUATION (DEPARTURE) A : CONNECTION POINT N : SHEET No. SH-TA- : SHEET OF "STEAM TURBINE GENERATOR AUXILIARY INTERLOCK DIAGRAM" SH-G- : SHEET OF THIS DRAWING SH-* : SAME SHEET

LOCATION KEY

LOCATION	EXPLANATION	SUPPLIER
GPP	GEN. PROTECTION RELAY PANEL	MHPS
GCP	GEN. CONTROL PANEL	MHPS
SYN	OPERATING CONSOLE OF SYNCHRONIZING (RACK MOUNTED)	MHPS
TCS	TURBINE CONTROL SYSTEM	MHPS
TCS HMI	TURBINE CONTROL SYSTEM HUMAN-MACHINE INTERFACE	MHPS
AVR	AUTOMATIC VOLTAGE REGULATOR CUBICLE	MHPS
VT/SA	VT & SA CUBICLE	MHPS
NGTR	GEN. NEUTRAL GROUNDING TRANSFORMER CUBICLE	MHPS
DCS	DISTRIBUTED CONTROL SYSTEM	OTHERS
DCS HMI	DISTRIBUTED CONTROL SYSTEM HUMAN-MACHINE INTERFACE	OTHERS
SOE	SEQUENCE OF EVENT RECORDER	OTHERS
GMCB	GEN. MAIN CIRCUIT BREAKER	OTHERS
RCP	RELAY CONTROL PANEL	OTHERS

SYMBOL LEGEND AND LOCATION KEY

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DEVICE LEGEND

DEVICE	EXPLANATION	SUPPLIER
3	CONTROL SWITCH	-
7	CONTROL SWITCH (ADJUSTMENT SWITCH)	-
41	FIELD SWITCH (FS)	-
43	SELECTOR SWITCH (CHANGE OVER SWITCH)	-
52	CIRCUIT BREAKER (CB)	-
89	DISCONNECTING SWITCH (DS)	-
52L BREAKER	230kV CB CONTROL SWITCH (CS : CLOSE/NEUTRAL/OPEN)	MHPS
52G BREAKER	52G CONTROL SWITCH (CS : CLOSE/NEUTRAL/OPEN)	MHPS
3-41	41 CONTROL PB (PB : CLOSE/OPEN)	MHPS
VOLTAGE CONTROL	GENERATOR VOLTAGE CONTROL SWITCH (CS : RAISE/LOWER)	MHPS
SPEED CONTROL	TURBINE SPEED CONTROL SWITCH (CS : RAISE/LOWER)	MHPS
43-25	SYNCHRONIZING OPERATION MODE CHANGEOVER SWITCH OF SYN (COS : MANUAL/OFF/AUTO)	MHPS
52G	GENERATOR CIRCUIT BREAKER	OTHERS
52L	230kV CIRCUIT BREAKER	OTHERS
ES	EARTHING SWITCH	-
DS	DISCONNECTING SWITCH	OTHERS
ASD1	AUTOMATIC SYNCHRONIZING DEVICE (FOR 52G)	MHPS
ASD2	AUTOMATIC SYNCHRONIZING DEVICE (FOR 52L)	MHPS
PSS	POWER SYSTEM STABILIZER	MHPS

DEVICE	EXPLANATION	SUPPLIER
ATS	AUTOMATIC TURBINE START-UP/SHUT-DOWN SYSTEM	MHPS
AVR	AUTOMATIC VOLTAGE REGULATOR	MHPS
GOV	GOVERNOR	MHPS
HMI	HUMAN-MACHINE INTERFACE	MHPS/OTHERS
I	INCOMING BUS OF SYNCHRONIZATION CIRCUIT FOR GENERATOR	MHPS
R	RUNNING BUS OF SYNCHRONIZATION CIRCUIT FOR GENERATOR	MHPS
SYN	OPERATING CONSOLE OF SYNCHRONIZING (RACK MOUNTED)	MHPS
25	SYNCHRO CHECK RELAY	MHPS
25A	SYNCHRO CHECK RELAY IN ASD	MHPS
25X1	AUX. RELAY FOR SYNCHRO OPERATION OF 52G	MHPS
25X2	AUX. RELAY FOR SYNCHRO OPERATION OF 52L	MHPS
25Y1	AUX. RELAY FOR CONNECTION OF ASD1	MHPS
25Y2	AUX. RELAY FOR CONNECTION OF ASD2	MHPS
15L	SPEED LOWER IN ASD	MHPS
15R	SPEED RAISE IN ASD	MHPS
60L	VOLTAGE LOWER IN ASD	MHPS
60R	VOLTAGE RAISE IN ASD	MHPS
84R	VOLTAGE RELAY (AUX. RELAY) FOR RUNNING BUS	MHPS
84GI	VOLTAGE RELAY (AUX. RELAY) FOR INCOMING BUS OF GEN. SIDE	MHPS
84LI	VOLTAGE RELAY (AUX. RELAY) FOR INCOMING BUS OF LINE SIDE	MHPS

DEVICE LEGEND

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OPERATION PHILOSOPHY

THE OPERATION PHILOSOPHY AND THE OPENING AND CLOSING LOGIC OF GENERATOR CIRCUIT BREAKER (52G), FIELD SWITCH (41) AND 230kV CIRCUIT BREAKER (52L) ARE INDICATED BELOW.

1. GENERATOR CIRCUIT BREAKER (52G)

1.1 OPERATING LOCATION

52G IS OPERATED BY THE FOLLOWING EQUIPMENT FROM THE FOLLOWING LOCATION.

OPERATION	OPERATION LOCATION	NO.	OPERATING SIGNAL	OPERATING EQUIPMENT	REMARKS
SELECTING	SYN	(A)	"AUTO/MANUAL/OFF"	43-25	MANUAL OPERATION
	SYN	-	"52G/52L"	52G, 52L SELECTOR	ALL OPERATION
	SYN	-	"SYN/REMOTE"	OPERATION LOCATION	ALL OPERATION
CLOSING	TCS	(B)	AUTO CLOSE	ATS	AUTOMATIC OPERATION
	DCS/SYN	(C)	AUTO SYNCHRONIZATION PB "START"	AUTO SYN. START PB	MANUAL OPERATION WITH ASD1
	SYN	(D)	52G CLOSE	52G BREAKER CS	MANUAL OPERATION WITHOUT ASD1
OPENING	SYN	(E)	52G OPEN	-	AUTOMATIC OPERATION
	SYN	(F)	52G OPEN	52G BREAKER CS	MANUAL OPERATION
	DCS	(G)	52G OPEN	52G BREAKER PB	MANUAL OPERATION

1.2 OPERATING CONCEPT OF 52G

1.2.1 SELECTING OPERATION OF 52G (AUTO/MANUAL/OFF)

A) OPERATOR SELECTS AUTO/MANUAL/OFF FROM SYN.

"AUTO" MODE CAN ACCEPT "ATS CLOSE ORDER OPERATION" AND "AUTOMATIC SYNCHRONIZING OPERATION.

"MANUAL" MODE CAN ACCEPT "MANUAL SYNCHRONIZING OPERATION".

"OFF" MODE CANNOT ACCEPT 52G CLOSING OPERATION.

1.2.2 CLOSING OPERATION OF 52G - (START UP)

1.2.2.1 ATS OPERATION

B) AUTOMATIC CLOSING OPERATION BY ATS

B-1) LOGIC VERIFIES "43-25" IS SET TO "AUTO" POSITION AT SYN.

B-2) 52G AUTOMATIC CLOSE OPERATION IS STARTED BY ATS ORDER.

1.2.2.2 AUTOMATIC SYNCHRONIZING OPERATION (DCS OR SYN)

C) CLOSING OPERATION WITH ASD1 BY MANUAL COMMAND

C-1) LOGIC VERIFIES "43-25" IS SET TO "AUTO" POSITION AT SYN.

C-2) OPERATOR PUSHES "AUTO SYN. START" PUSHBUTTON TO ENERGIZE THE SYNCHRONIZATION CIRCUIT AND TO START ASD1.

C-3) ASD1 OUTPUTS THE "RAISE / LOWER" COMMAND TO AVR AND TCS TO ADJUST GENERATOR VOLTAGE AND TURBINE SPEED.

C-4) ASD1 OUTPUTS THE "CLOSE" COMMAND TO 52G WHEN THE ADJUSTED VOLTAGE AND SPEED GO INTO THE PRE-DETERMINED RANGE.

C-5) 52G IS CLOSED BY THE COMMAND FROM ASD1.

1.2.2.3 MANUAL SYNCHRONIZING OPERATION (SYN)

D) CLOSING OPERATION WITHOUT ASD1 BY OPERATOR

D-1) LOGIC VERIFIES "43-25" IS SET TO "MANUAL" POSITION AT SYN.

D-2) THE OPERATOR VERIFIES THE VOLTAGE, FREQUENCY AND PHASE DEVIATION BETWEEN GENERATOR AND LOW VOLTAGE SIDE BUS OF GEN. TRANSF. WITH VOLTMETER, FREQUENCY METER AND SYNCHROSCOPE AT SYN AND ADJUSTS GENERATOR VOLTAGE AND TURBINE SPEED WITH CONTROL SWITCHES OF VOLTAGE CONTROL AND SPEED CONTROL AT SYN.

D-3) OPERATOR VERIFIES THE ADJUSTED VOLTAGE, ADJUSTED SPEED AND PHASE GO INTO THE PRE-DETERMINED RANGE.

D-4) OPERATOR TURNS 52G BREAKER TO "CLOSE" POSITION AT SYN AND 52G IS CLOSED.

1.2.3 OPENING OPERATION

E) AUTOMATIC OPENING OPERATION

E-1) 52G IS OPENED BY LOCKOUT RELAY OF GEN., GEN. TRANSF. AND UNIT TRANSF. OR ST TRIP CONDITION UNDER 52G CLOSED.

F) OPENING OPERATION BY OPERATOR (SYN)

F-1) OPERATOR TURNS 52G BREAKER TO "OPEN" POSITION AT SYN.

F-2) 52G IS OPENED BY THE OPERATOR ACTION FROM SYN.

G) OPENING OPERATION BY OPERATOR (DCS)

G-1) OPERATOR PUSHES "52G OPEN" PUSHBUTTON AND 52G WILL OPEN.

1.3 INTERLOCK CONDITION

THE FOLLOWING INTERLOCK CONDITIONS ARE IMPLEMENTED IN 52G CLOSING LOGIC.

1) GEN., GEN. TRANSF. AND UNIT TRANSF. PROTECTION RELAY ARE NOT OPERATED CONDITIONS BEFORE 52G IS OPERATED TO CLOSE.

2) 52G CAN BE CLOSED ONLY WHILE THE SYNCHRONIZING CHECK RELAY (25) IS OPERATED.

3) FIELD SWITCH (41) AND 52L ARE CLOSED.

2. FIELD SWITCH (41)

2.1 OPERATING LOCATION

41 IS OPERATED BY THE FOLLOWING EQUIPMENT FROM THE FOLLOWING LOCATION.

OPERATION	OPERATION LOCATION	NO.	OPERATING SIGNAL	OPERATING EQUIPMENT	REMARKS
CLOSING	TCS	(A)	AUTO CLOSE	ATS	AUTOMATIC OPERATION
	TCS	(B)	41 CLOSE	3-41	MANUAL OPERATION
OPENING	TCS	(C)	41 OPEN	-	AUTOMATIC OPERATION
	TCS	(D)	41 OPEN	3-41	MANUAL OPERATION

2.2 OPERATING CONCEPT OF 41

2.2.1 CLOSING OPERATION OF 41

A) AUTOMATIC CLOSING OPERATION BY ATS

A-1) TCS OUTPUTS THE "EXCITATION AUTO START SIGNAL" TO 41 IN AVR UNDER TURBINE FULL SPEED / NO-LOAD CONDITION.

A-2) 41 IS CLOSED BY THE COMMAND FROM ATS.

B) MANUAL CLOSING OPERATION BY OPERATOR (TCS)

B-1) OPERATOR PUSHES "3-41 CLOSE" PUSHBUTTON AND 41 WILL CLOSE.

2.2.2 OPENING OPERATION OF 41

C) AUTOMATIC OPENING OPERATION

C-1) 41 IS OPENED BY LOCKOUT RELAY OF GEN., GEN. TRANSF. AND UNIT TRANSF. OR ST TRIP CONDITION UNDER 52G OPEN.

D) MANUAL OPENING OPERATION BY THE OPERATOR (TCS)

D-1) LOGIC VERIFIES 52G IS OPENED.

D-2) OPERATOR PUSHES "3-41 OPEN" PUSHBUTTON AND 41 WILL OPEN.

2.3 INTERLOCK CONDITION

2.3.1 THE FOLLOWING INTERLOCK CONDITIONS ARE IMPLEMENTED IN 41 CLOSING LOGIC.

1) TURBINE SPEED IS OVER (LATER)rpm SPEED BEFORE 41 IS OPERATED TO CLOSE.

2) 52G AND ES ARE OPENED BEFORE 41 IS OPERATED TO CLOSE.

3) GEN. PROTECTION RELAY ARE NOT OPERATED CONDITIONS BEFORE 41 IS OPERATED TO CLOSE.

2.3.2 THE FOLLOWING INTERLOCK CONDITION IS IMPLEMENTED IN 41 OPENING LOGIC.

1) 52G IS OPENED BEFORE 41 IS OPERATED TO OPEN.

OPERATION PHILOSOPHY

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3. 230kV CIRCUIT BREAKER (52L)

3.1 OPERATING LOCATION

230kV CB IS OPERATED BY THE FOLLOWING EQUIPMENT FROM THE FOLLOWING LOCATION.

OPERATION	OPERATION LOCATION	NO.	OPERATING SIGNAL	OPERATING EQUIPMENT	REMARKS
SELECTING	SYN	(A)	"AUTO/MANUAL/OFF"	43-25	MANUAL OPERATION
	SYN	-	"52G/52L"	52G, 52L SELECTOR	ALL OPERATION
	SYN	-	"SYN/REMOTE"	OPERATION LOCATION	ALL OPERATION
CLOSING	DCS/SYN	(B)	AUTO SYNCHRONIZATION PB "START"	AUTO SYN. START PB	MANUAL OPERATION WITH ASD2
	SYN	(C)	52L CLOSE	52L BREAKER CS	MANUAL OPERATION WITHOUT ASD2
	SYN	(D)	52L CLOSE	52L BREAKER CS	MANUAL OPERATION FOR FIRST POWER RECEIVE
OPENING	SYN	(E)	52L OPEN	-	AUTOMATIC OPERATION
	SYN	(F)	52L OPEN	52L BREAKER CS	MANUAL OPERATION
	DCS	(G)	52L OPEN	52L BREAKER PB	MANUAL OPERATION

3.2 OPERATING CONCEPT OF 52L

3.2.1 SELECTING OPERATION OF 52L (AUTO/MANUAL/OFF)

A) OPERATOR SELECTS AUTO/MANUAL/OFF FROM SYN.
"AUTO" MODE CAN ACCEPT "ATS CLOSE ORDER OPERATION" AND "AUTOMATIC SYNCHRONIZING OPERATION.
"MANUAL" MODE CAN ACCEPT "MANUAL SYNCHRONIZING OPERATION".
"OFF" MODE CANNOT ACCEPT 52L CLOSING OPERATION.

3.2.2 CLOSING OPERATION OF 52L - (START UP)

3.2.2.1 AUTOMATIC SYNCHRONIZING OPERATION (DCS OR SYN)

B) CLOSING OPERATION WITH ASD2 BY MANUAL COMMAND
B-1) LOGIC VERIFIES "43-25" IS SET TO "AUTO" POSITION AT SYN.
B-2) OPERATOR PUSHES "AUTO SYN START" PUSHBUTTON TO ENERGIZE THE SYNCHRONIZATION CIRCUIT AND TO START ASD2.
B-3) ASD2 OUTPUTS THE "RAISE / LOWER" COMMAND TO AVR AND TCS TO ADJUST GENERATOR VOLTAGE AND TURBINE SPEED.
B-4) ASD2 OUTPUTS THE "CLOSE" COMMAND TO 52L WHEN THE ADJUSTED VOLTAGE AND SPEED GO INTO THE PRE-DETERMINED RANGE.
B-5) 52L IS CLOSED BY THE COMMAND FROM ASD2.

3.2.2.2 MANUAL SYNCHRONIZING OPERATION (SYN)

C) CLOSING OPERATION WITHOUT ASD2 BY OPERATOR
C-1) LOGIC VERIFIES "43-25" IS SET TO "MANUAL" POSITION AT SYN.
C-2) THE OPERATOR VERIFIES THE VOLTAGE, FREQUENCY AND PHASE DEVIATION BETWEEN GENERATOR AND LOW VOLTAGE SIDE BUS OF GEN. TRANSF. WITH VOLTMETER, FREQUENCY METER AND SYNCHROSCOPE AT SYN AND ADJUSTS GENERATOR VOLTAGE AND TURBINE SPEED WITH CONTROL SWITCHES OF VOLTAGE CONTROL AND SPEED CONTROL AT SYN.
C-3) OPERATOR VERIFIES THE ADJUSTED VOLTAGE, ADJUSTED SPEED AND PHASE GO INTO THE PRE-DETERMINED RANGE.
C-4) OPERATOR TURNS 52L BREAKER TO "CLOSE" POSITION AT SYN AND 52L IS CLOSED.

3.2.2.3 MANUAL CLOSE FOR FIRST POWER RECEIVE (SYN)

D) CLOSING OPERATION WITHOUT SYNCHRONIZING
D-1) LOGIC VERIFIES "43-25" IS SET TO "MANUAL" POSITION AT SYN.
D-2) THE OPERATOR VERIFIES "52L CLOSE PERMIT CONDITION FOR FIRST POWER RECEIVE".
D-3) OPERATOR TURNS 52L BREAKER TO "CLOSE" POSITION AT SYN AND 52L IS CLOSED.

3.2.3 OPENING OPERATION

E) AUTOMATIC OPENING OPERATION
E-1) 52L IS OPENED BY LOCKOUT RELAY OF 230kV SWYD., GEN. TRANSF. AND UNIT TRANSF. UNDER 52L CLOSED.

F) OPENING OPERATION BY OPERATOR (SYN)
F-1) OPERATOR TURNS 52L BREAKER TO "OPEN" POSITION AT SYN.
F-2) 52L IS OPENED BY THE OPERATOR ACTION FROM SYN.

G) OPENING OPERATION BY OPERATOR (DCS)
G-1) OPERATOR PUSHES "52L OPEN" PUSHBUTTON AND 52L WILL OPEN.

3.2.4 INTERLOCK CONDITION

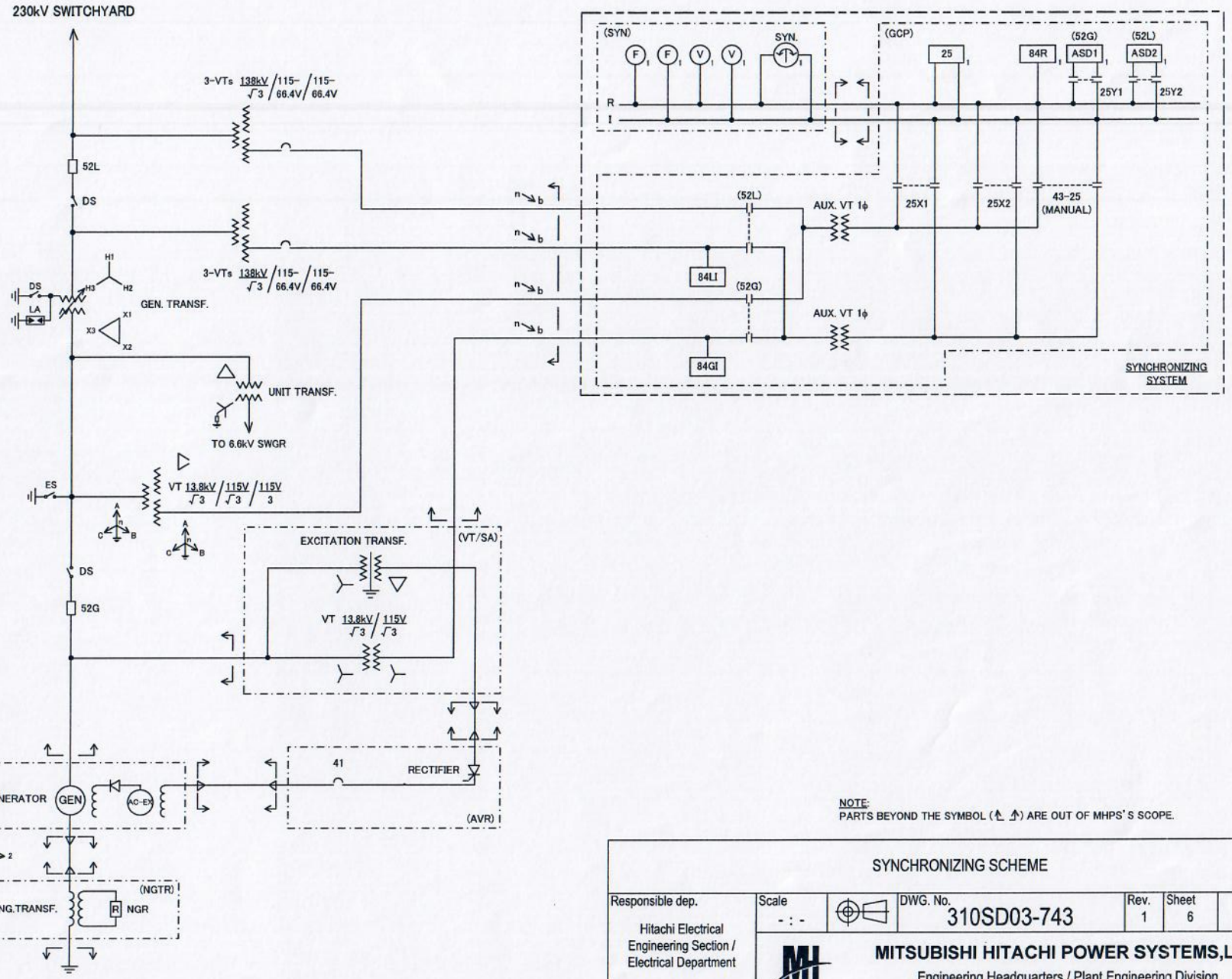
THE FOLLOWING INTERLOCK CONDITIONS ARE IMPLEMENTED IN 52L CLOSING LOGIC.
1) GEN., GEN. TRANSF. AND UNIT TRANSF. PROTECTION RELAY ARE NOT OPERATED CONDITIONS BEFORE 52L IS OPERATED TO CLOSE.
2) 52L CAN BE CLOSED ONLY WHILE THE SYNCHRONIZING CHECK RELAY (25) IS OPERATED WITHOUT FIRST POWER RECEIVE.
3) 52G IS CLOSED WITHOUT FIRST POWER RECEIVE.

OPERATION PHILOSOPY

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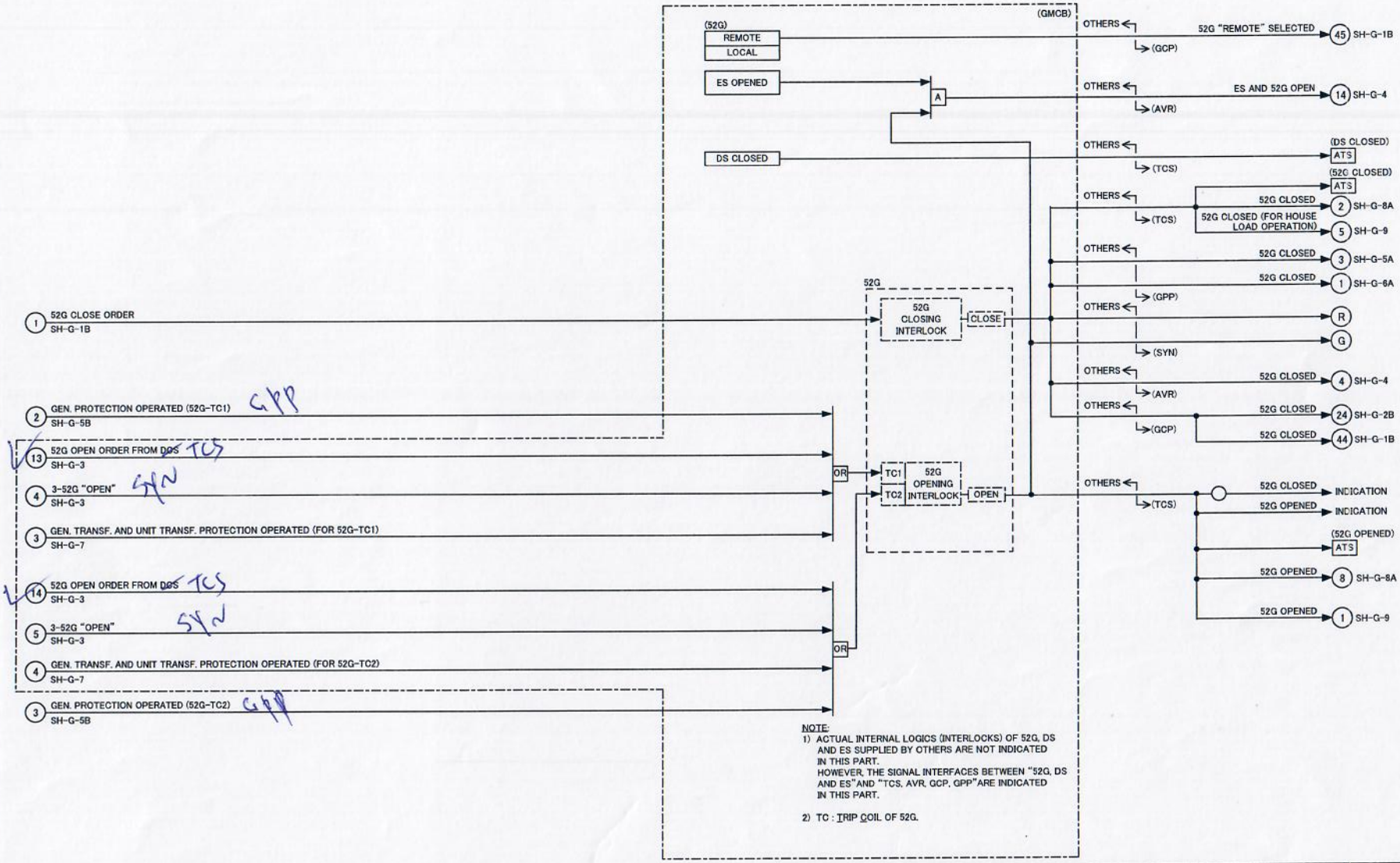
NOTE:
PARTS BEYOND THE SYMBOL (↑ ↑) ARE OUT OF MHPS'S SCOPE.

SYNCHRONIZING SCHEME

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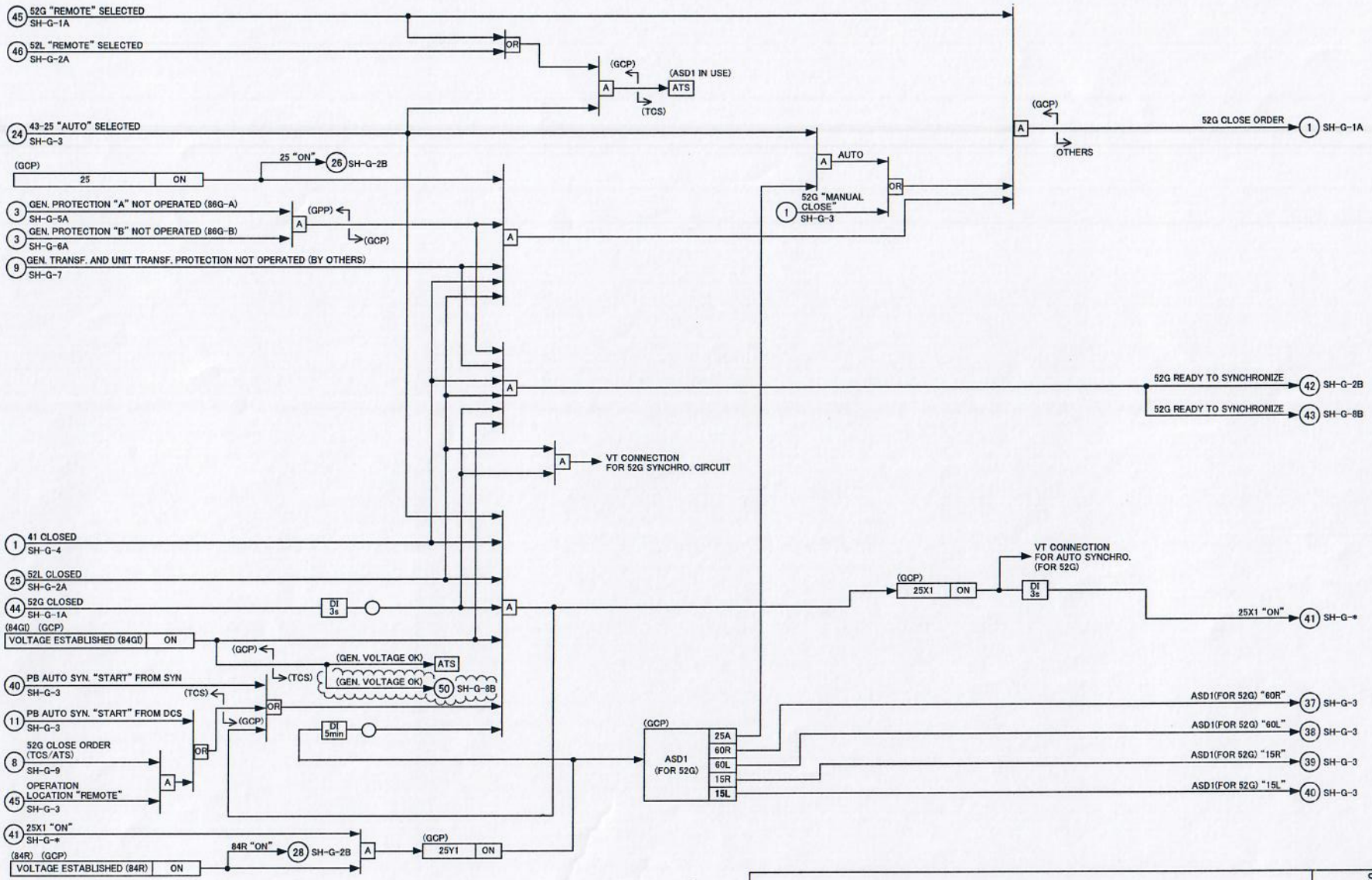
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GENERATOR CIRCUIT BREAKER LOGIC (52G) (1/2)					SH. No.
					SH-G-1A
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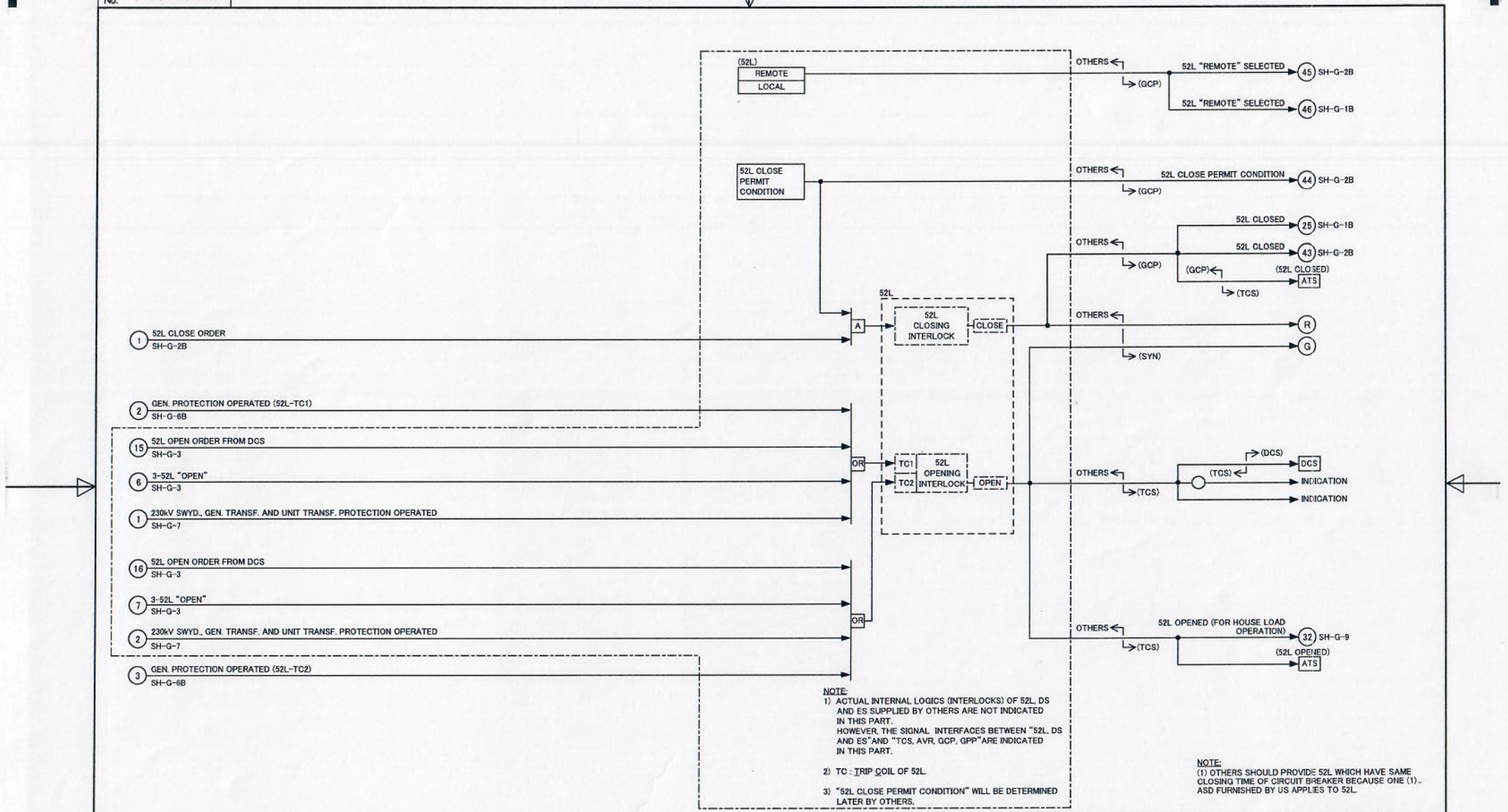
DWG. No. 310SD03-743



GENERATOR CIRCUIT BREAKER LOGIC (52G) (2/2)					SH. No. SH-G-1B
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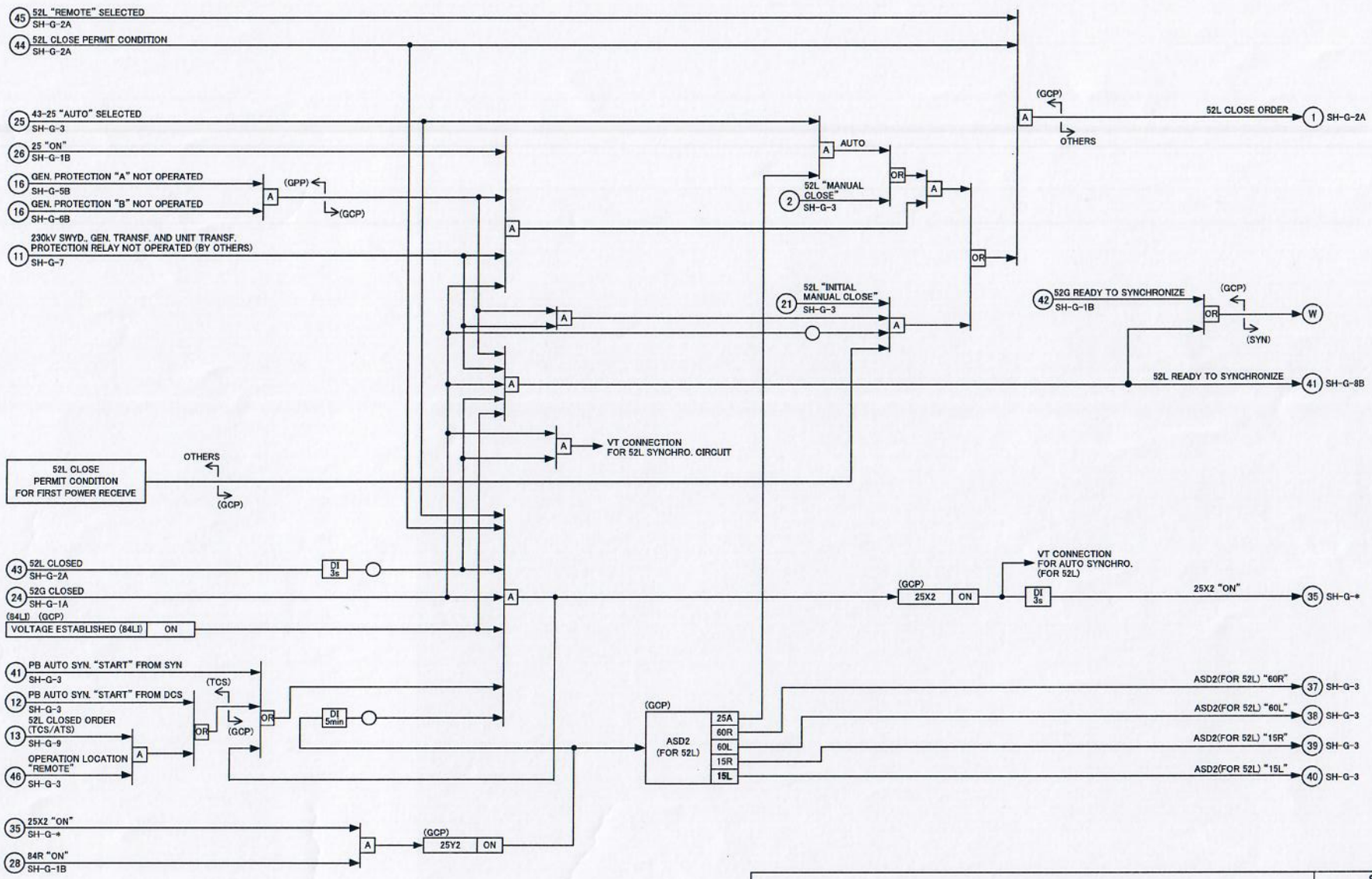
NOTE:
1) ACTUAL INTERNAL LOGICS (INTERLOCKS) OF 52L, DS AND ES SUPPLIED BY OTHERS ARE NOT INDICATED IN THIS PART.
HOWEVER, THE SIGNAL INTERFACES BETWEEN "52L, DS AND ES" AND "TCS, AVR, GCP, GPP" ARE INDICATED IN THIS PART.
2) TC : TRIP COIL OF 52L.
3) "52L CLOSE PERMIT CONDITION" WILL BE DETERMINED LATER BY OTHERS.

NOTE:
(1) OTHERS SHOULD PROVIDE 52L WHICH HAVE SAME CLOSING TIME OF CIRCUIT BREAKER BECAUSE ONE (1) ASD FURNISHED BY US APPLIES TO 52L.

230kV CIRCUIT BREAKER LOGIC (52L) (1/2)					SH. No. SH-G-2A
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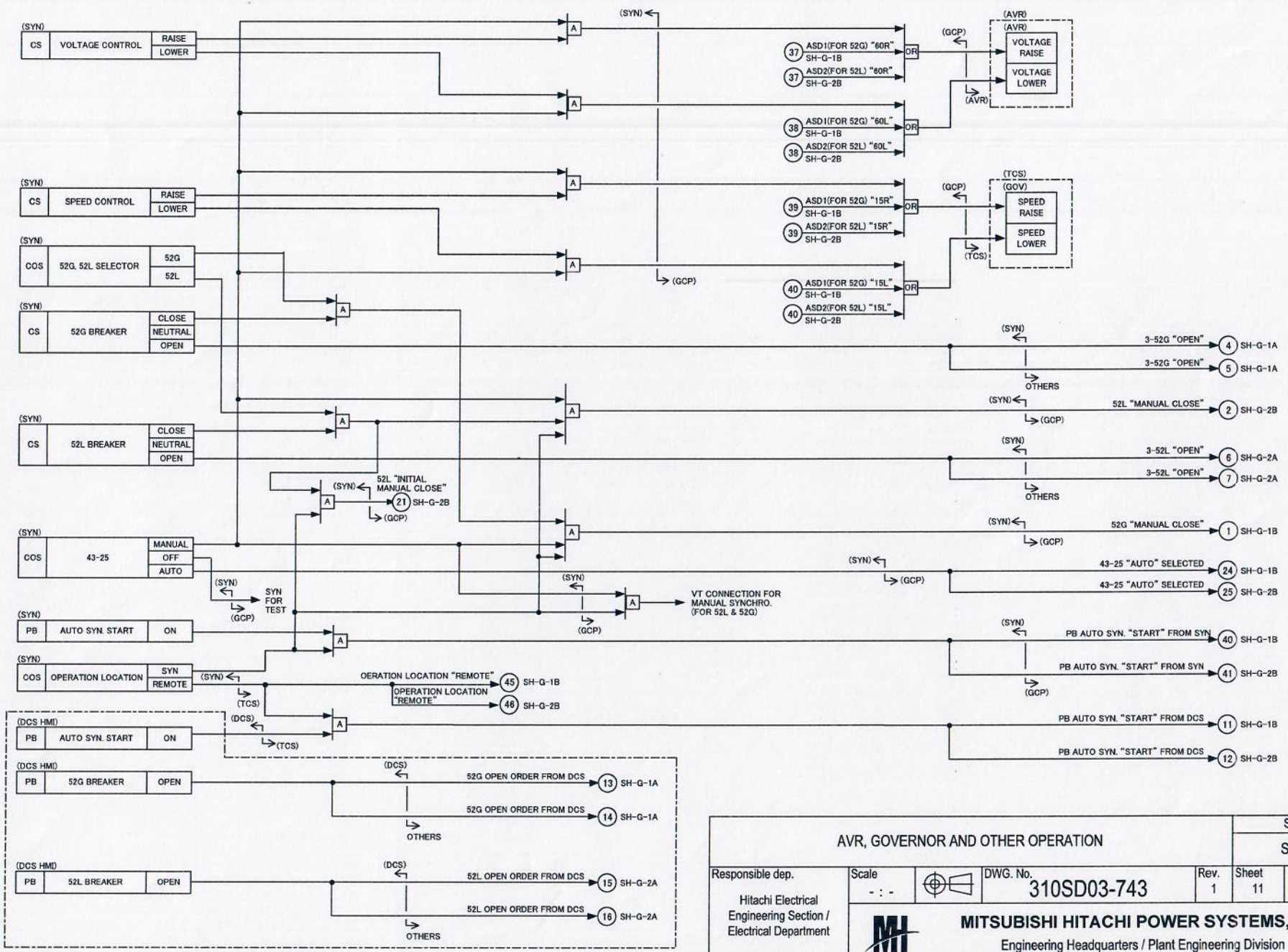
DWG. No. 310SD03-743



230kV CIRCUIT BREAKER LOGIC (52L) (2/2)					SH. No. SH-G-2B
Responsible dep. Hitachi Electrical Engineering Section / Electrical Department	Scale - : -		DWG. No. 310SD03-743	Rev. 1	Sheet 10

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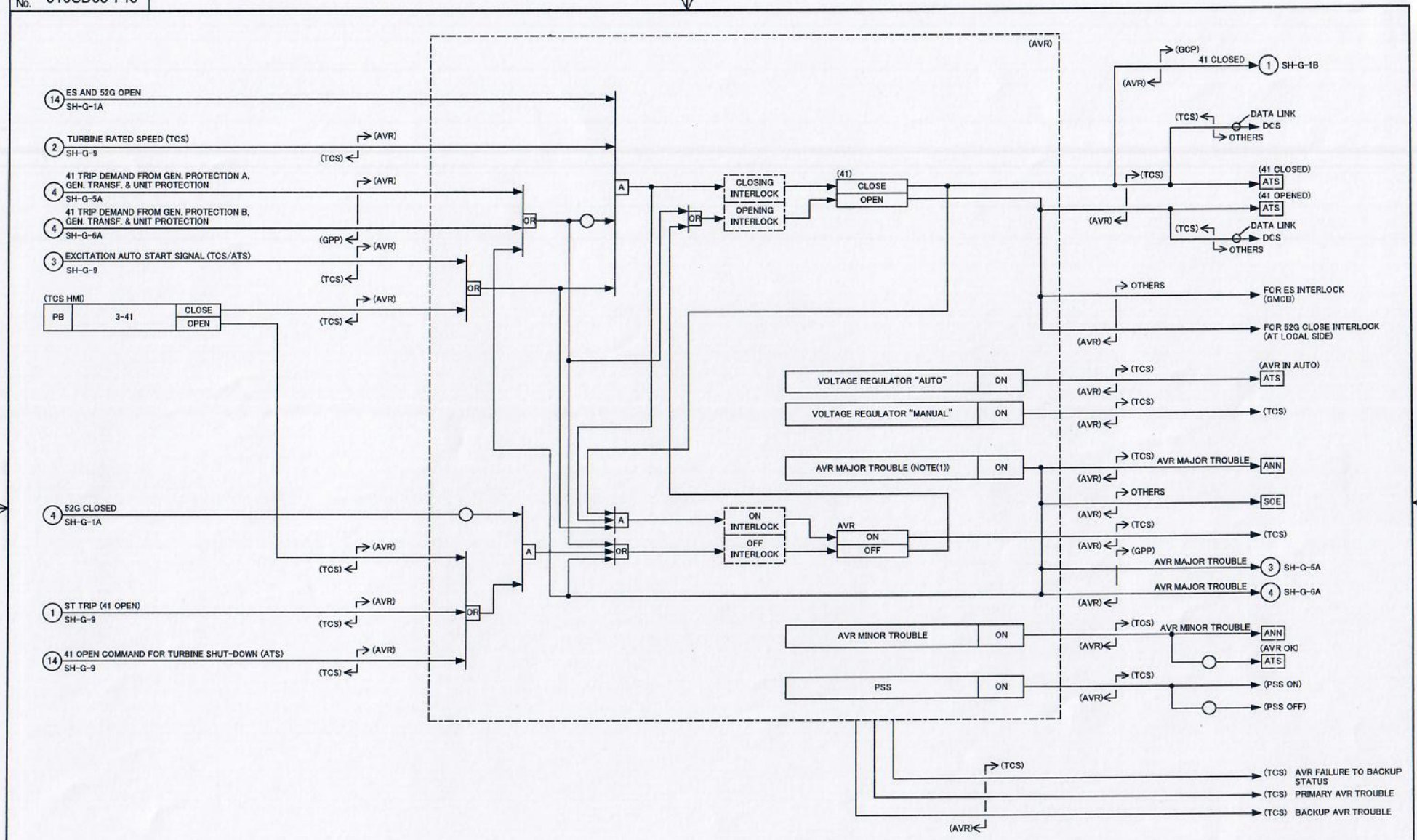
AVR, GOVERNOR AND OTHER OPERATION

SH. No. SH-G-3

Responsible dep.	Scale	DWG. No.	Rev.	Sheet
Hitachi Electrical Engineering Section / Electrical Department	- : -	310SD03-743	1	11

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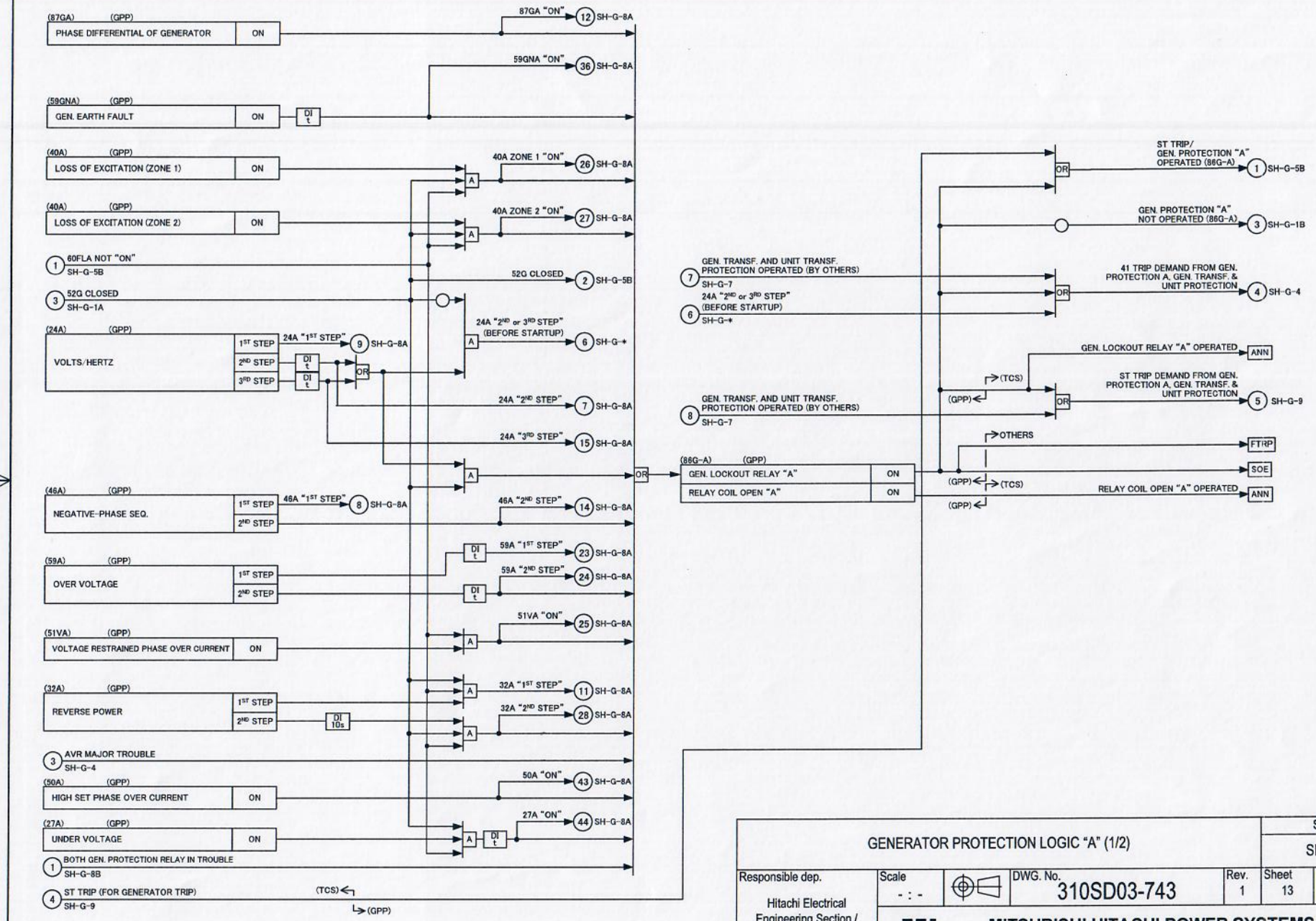


NOTE
(1) MAIN TRIP ITEMS OF "AVR MAJOR TROUBLE" ARE AS FOLLOWS.
- FIELD CIRCUIT OVERCURRENT PROTECTION (INSTANTANEOUS / INVERSE TIME)
- VOLTS PER HERTZ PROTECTION
- LOSS OF EXCITATION PROTECTION
- TWO CONVERTER FAILURE
- CROWBAR FAILURE
- FIELD FLASH. FAULT

FIELD SWITCH (41) LOGIC				SH. No. SH-G-4
Responsible dep. Hitachi Electrical Engineering Section / Electrical Department	Scale - : -		DWG. No. 310SD03-743	Rev. 1 / Sheet 12

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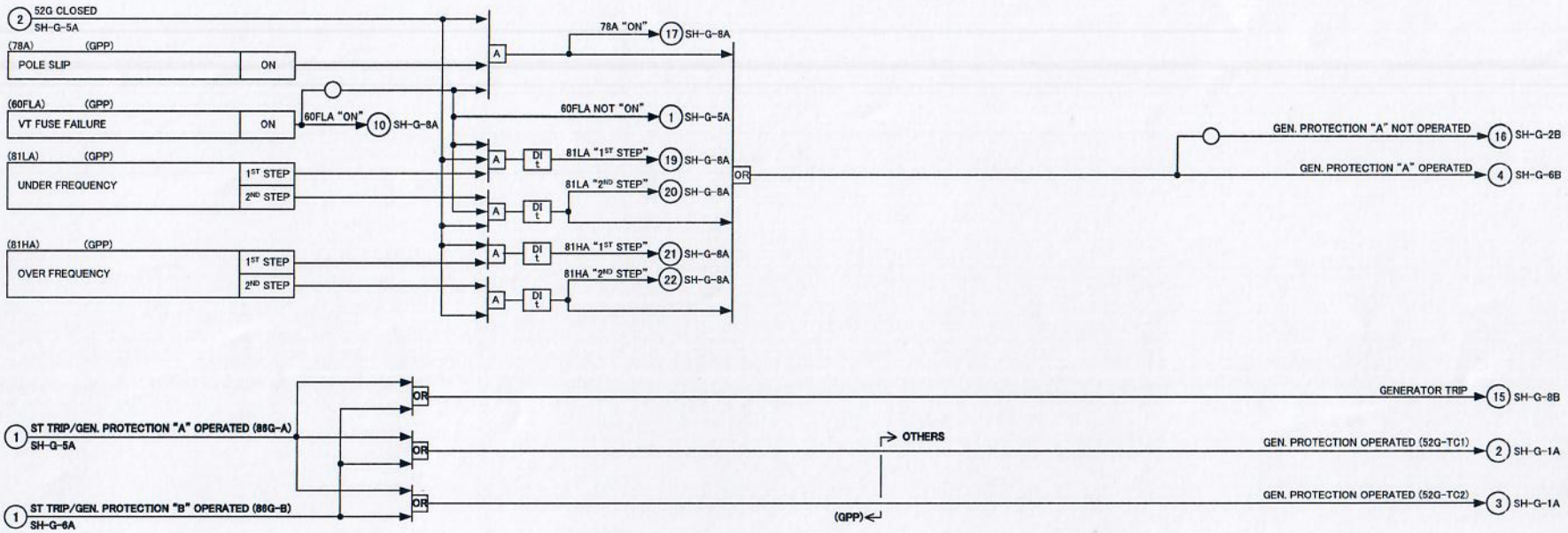
GENERATOR PROTECTION LOGIC "A" (1/2)

SH. No. SH-G-5A

Responsible dep.	Scale	DWG. No.	Rev.	Sheet
Hitachi Electrical Engineering Section / Electrical Department	- : -	310SD03-743	1	13

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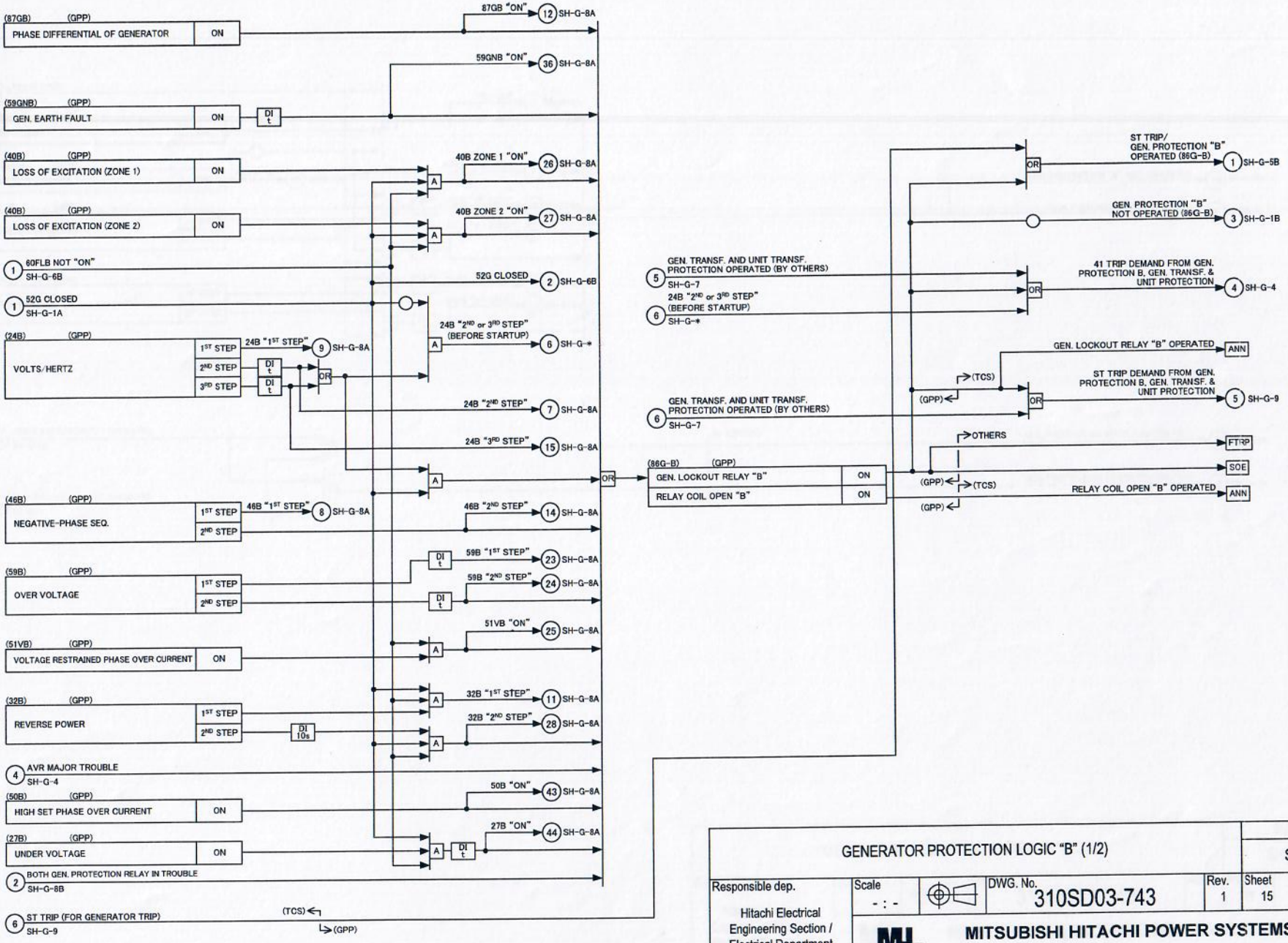
DWG. No. 310SD03-743



GENERATOR PROTECTION LOGIC "A" (2/2)					SH. No. SH-G-5B
Responsible dep. Hitachi Electrical Engineering Section / Electrical Department	Scale - : -		DWG. No. 310SD03-743	Rev. 1	Sheet 14

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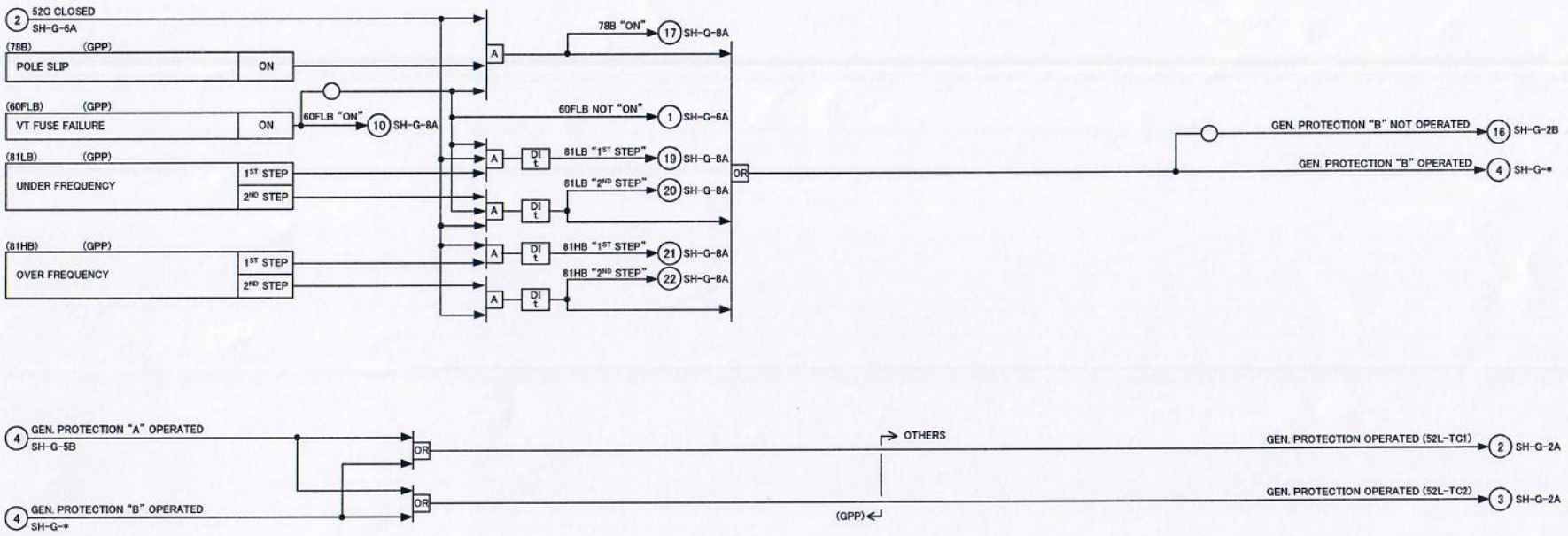
GENERATOR PROTECTION LOGIC "B" (1/2)

SH. No. SH-G-6A

Responsible dep.	Scale	DWG. No.	Rev.	Sheet
Hitachi Electrical Engineering Section / Electrical Department	- : -	310SD03-743	1	15

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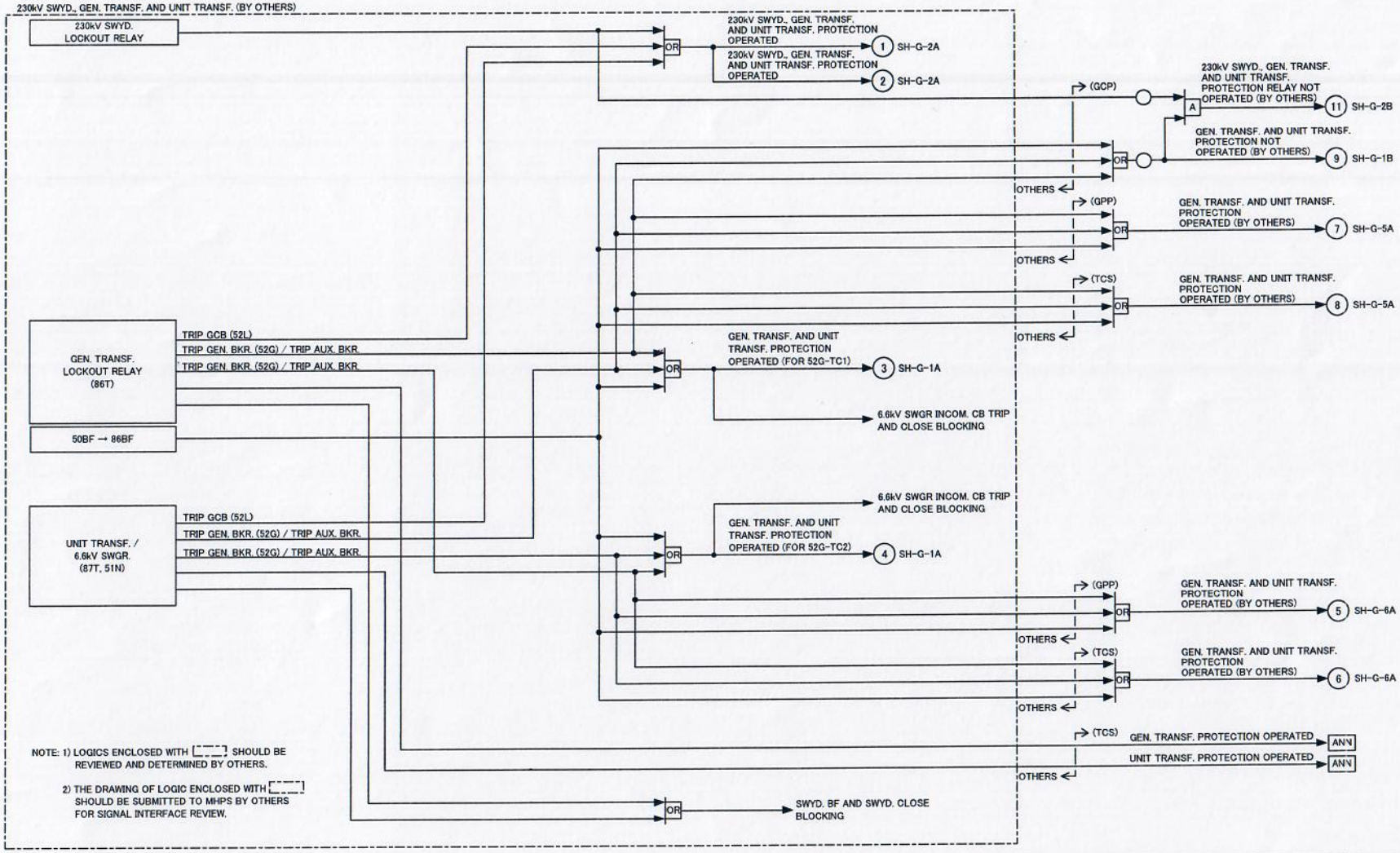
DWG. No. 310SD03-743



GENERATOR PROTECTION LOGIC "B" (2/2)					SH. No. SH-G-6B
Responsible dep. Hitachi Electrical Engineering Section / Electrical Department	Scale - : -		DWG. No. 310SD03-743	Rev. 1	Sheet 16

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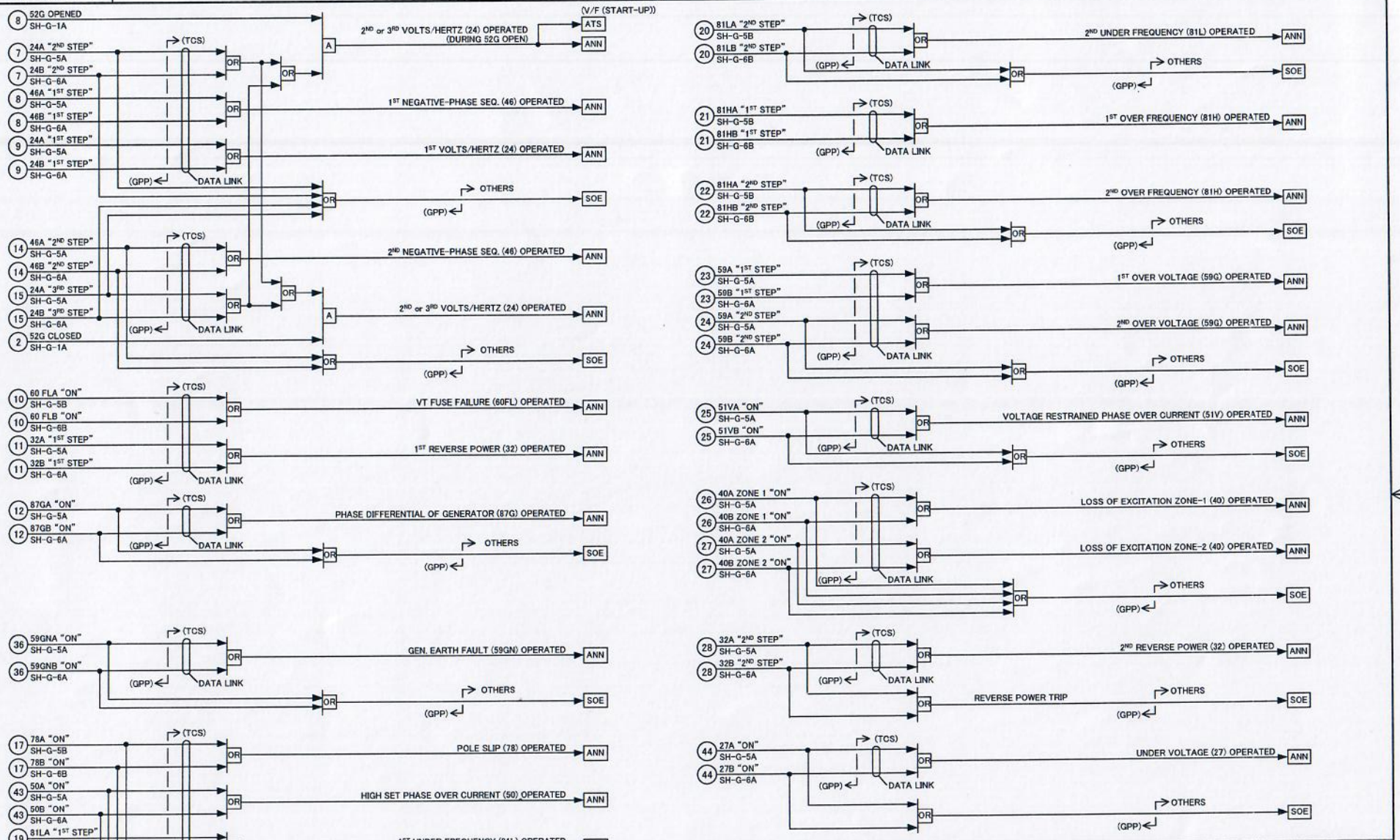
DWG. No. 310SD03-743



230kV SWITCHYARD AND TRANSFORMER PROTECTION LOGIC					SH. No. SH-G-7
Responsible dep. Hitachi Electrical Engineering Section / Electrical Department	Scale - : -		DWG. No. 310SD03-743	Rev. 1	Sheet 17

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DWG. No. 310SD03-743



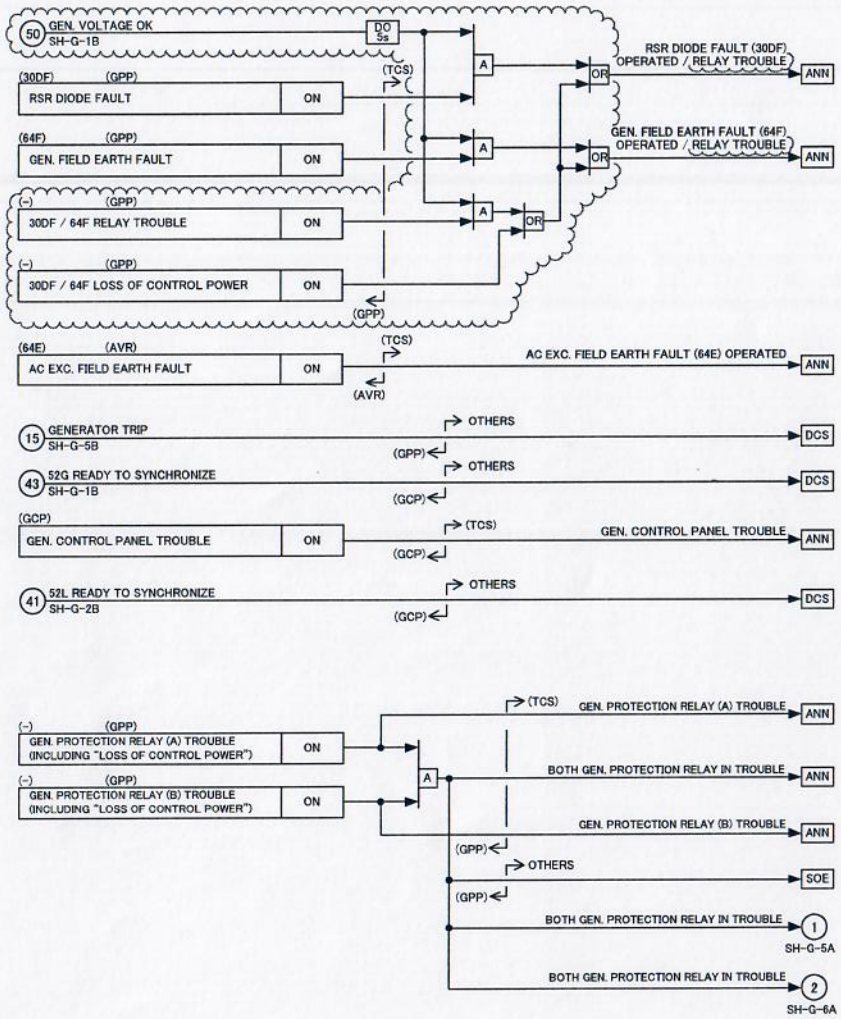
ALARM FOR GENERATOR PROTECTION (1/2)

SH. No. SH-G-8A

Responsible dep.	Scale	DWG. No.	Rev.	Sheet
Hitachi Electrical Engineering Section / Electrical Department	- : -	310SD03-743	1	18

MITSUBISHI HITACHI POWER SYSTEMS,LTD.
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DWG. No. 310SD03-743



ALARM FOR GENERATOR PROTECTION (2/2)					SH. No. SH-G-8B
Responsible dep. Hitachi Electrical Engineering Section / Electrical Department	Scale - : -		DWG. No. 310SD03-743	Rev. 1	Sheet 19

MITSUBISHI HITACHI POWER SYSTEMS,LTD.
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DWG. No. 310SD03-743

(1) 52G OPENED SH-G-1A	→	(12)	SH-TC-2 (310SC90-774)
(–) 41 OPEN COMMAND FOR TURBINE SHUT-DOWN (ATS) (310SC90-775, SH-30)	→	(14)	SH-G-4
(2) ST TRIP (41 OPEN) (310SC90-774, SH-TA-1)	→	(1)	SH-G-4
(1) TURBINE RATED SPEED (TCS) (310SC90-774, SH-TD-1)	→	(2)	SH-G-4
(–) EXCITATION AUTO START SIGNAL (TCS/ATS) (310SC90-775, SH-24)	→	(3)	SH-G-4
(1) ST TRIP (FOR GENERATOR TRIP) (310SC90-774, SH-TA-6)	→	(4)	SH-G-5A
(2) ST TRIP (FOR GENERATOR TRIP) (310SC90-774, SH-TA-6)	→	(6)	SH-G-6A
(–) 52G CLOSE ORDER (TCS/ATS) (310SC90-775, SH-25)	→	(8)	SH-G-1B
(32) 52L OPENED (FOR HOUSE LOAD OPERATION) SH-G-2A	→	(15)	SH-TD-10 (310SC90-774)
(5) ST TRIP DEMAND FROM GEN. PROTECTION A, GEN. TRANSF. & UNIT PROTECTION SH-G-5A	→	(9)	SH-TA-9 (310SC90-774)
(5) ST TRIP DEMAND FROM GEN. PROTECTION B, GEN. TRANSF. & UNIT PROTECTION SH-G-6A	→	(10)	SH-TA-9 (310SC90-774)
(5) 52G CLOSED (FOR HOUSE LOAD OPERATION) SH-G-1A	→	(11)	SH-TA-6 (310SC90-774)
(–) 52L CLOSED ORDER (TCS/ATS) (310SC90-775, SH-25)	→	(13)	SH-G-2B

TURBINE OPERATION

SH. No. SH-G-9

Responsible dep.	Scale	DWG. No.	Rev.	Sheet
Hitachi Electrical Engineering Section / Electrical Department	- : -	310SD03-743	1	20

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